Gene Expression: a look at cats

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Genetics and environment



The environment in which an organism lives can affect how its genetic information is expressed. The Siamese cat has a pale coat arising from a mutation in the pigmentation gene. Pigmentation produces an organism's colour. The Siamese mutation (selected for by breeders because of the distinct appearance it produces) results in the chocolate colour only being expressed at the coolest parts of the body. On cats, these areas are the face, ears, paws and tail.

1 Give another example of how an organism's environment can affect its phenotype.

2 Some animals (including humans) have defective pigmentation genes, producing white hair, pale skin and red eyes with a blue or pink iris. Name this condition. _____

Random X-inactivation

Another easily recognisable cat is the Calico cat, which has a 'patchwork' coat of yellow, black and white. With the exception of identical twins, no two Calicos will be alike because their colouration is randomly decided during early development. Fur colour on cats is determined by genes on the X chromosome and the patchwork results from a process called *random X-inactivation*. As the name suggests, certain instructions on the X chromosome are inactivated at random.

Female cats receive two X chromosomes (males receive one). In Calico cats, one chromosome will contain the black gene (X^b) and the other contains the (alternative) yellow gene (X^y). As cells divide during development, one X chromosome in each cell will be inactivated at random. Daughter cells arising from the original parent cell will possess instructions to grow fur of the remaining colour. Sometimes the genes are instructed not to produce any colour and this is where the white fur appears. (Note: linkages to other genes are involved in this.)

3 Is it possible to produce a breed of Calico cat in which individuals are almost identical? Explain your answer.

4 Yellow 'Marmalade' cats are more commonly male than female. Why?

5 'Mr Big' was an enormous black and yellow cat. Was this cat male or female? Explain.

6 Is it possible to have male cats that are black and white, or yellow and white? Explain.

7 Why might cat breeders try to change the appearance of cats? _____

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